

Arush Jain

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Summary

B.Tech student specializing in Smart Manufacturing with a strong foundation in Machine Learning, Statistics, and Data Structures & Algorithms. Experienced in designing, training, and deploying end-to-end ML systems — including probabilistic demand-forecasting engines with conformal prediction guarantees and deep learning recommender systems with FAISS-based retrieval — using Python, PyTorch, and LightGBM, complemented by full-stack engineering skills (React.js, Next.js, Node.js) for shipping production-grade APIs. Proven ability to independently architect, benchmark, and ship ML-driven products, backed by rigorous experimentation and ablation studies. Seeking to apply strong ML fundamentals and software engineering skills to build reliable, production-grade machine learning systems.

Education

IIITDM Jabalpur

2023 – Present

B.Tech in Smart Manufacturing

Technical Skills

Languages: Python, JavaScript, TypeScript, C++

Machine Learning: PyTorch, LightGBM, scikit-learn, Pandas, NumPy, Conformal Prediction, Quantile Regression, Contrastive Learning (InfoNCE), Recommender Systems, Time-Series Forecasting

MLOps & Deployment: FastAPI, Docker, FAISS (Vector Search), CI/CD (GitHub Actions), Google Cloud Platform, AWS

Web Development: React.js, Next.js, Node.js, Express.js, REST APIs, Tailwind CSS

Databases: MongoDB, PostgreSQL, Supabase

Tools: Git, GitHub, Postman, Figma

Core CS: Data Structures & Algorithms, Probability & Statistics, Object-Oriented Programming, System Design

Projects

Conformal Demand Forecasting

Python, LightGBM, FastAPI, Docker

Probabilistic Demand Forecasting & Risk-Bounded Inventory Replenishment

- Built a probabilistic demand-forecasting engine using multi-quantile LightGBM and Conformalized Quantile Regression (CQR) to produce distribution-free prediction intervals with finite-sample coverage guarantees.
- Mapped calibrated 90% intervals to optimal order quantities via Newsvendor critical-fractile optimization, cutting stockout rate from 48% to 11% and reducing total inventory cost by 31% versus a point-forecast baseline across 3,680 held-out test items.
- Engineered a leak-free train/calibration/test time-series split with lag and rolling-window features, and shipped the model as a containerized FastAPI microservice validated by a pytest suite and CI pipeline (GitHub Actions).

Multimodal Two-Tower Product Recommender

PyTorch, FAISS, Python

Deep Learning Candidate Retrieval System with Hard-Negative Mining

- Designed and trained a two-tower dual-encoder retrieval model (user/item MLP towers with FAISS similarity search), benchmarked on a MovieLens warm-start split spanning 55K+ users and 20K+ candidate items.
- Implemented a hard-negative InfoNCE contrastive training loop with a custom FAISS-based hard-negative sampler, improving Recall@10 by 4.2% and Recall@50 by 3.6% over an in-batch-negative baseline.
- Ran a controlled 6-model ablation study (text features, gated fusion, Stage-2 MLP reranking) to isolate architectural impact, backed by a 53-test pytest suite guaranteeing zero train/test leakage.

Patent

Automatic Footrest Assembly for Two Wheeler

2025

- Designed a sensor-based automation system using embedded controllers and pressure sensors.
- Implemented hardware-software integration workflows with fail-safe operational logic.

Achievements

- **Google Cloud Arcade Facilitator Program** - deployed cloud-native applications on GCP using Compute Engine and Cloud Run.
- Selected for **Amazon ML Summer School 2026 (AMLSS)**, a competitive program focused on Machine Learning and Applied Data Science.

Co-Curricular Activities

- Organized and managed College Literature Club events, coordinating execution, logistics, and team workflows.
- Active member of a community Run Club, co-organizing running events and group games.